Amar Al-Zubaidi

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EDUCATION

Computer Science, BSc, Hons, Co-op — September 2024 – April 2029 (expected)
York University — North York, Ontario, Canada

- **Cumulative GPA:** 3.7/4.0 (7.85/9.00 institutional scale), member of the Dean's Honour Roll
- **Coursework:** Data Structures & Algorithms, Software Design, Computer Organization, Numerical Methods

PROJECTS

Comicpress – `github.com/amarz45/comicpress` — September 2025 – Present

- Built a **C++26/Qt** comic compression app for e-readers, published on Flathub for **Linux** with **100+ monthly downloads** and as an auto-updating **Windows** installer built by GitHub Actions with Azure Artifact Signing
- Cuts file sizes by **up to 80%** while *improving* on-device image quality by tailoring colour space, quantization, dithering, and resampling to each target device, discarding only data the device cannot display
- **Parallelized** the libvips-based image processing pipeline using worker **processes** instead of **threads** to work around PDFium's lack of thread safety, achieving **near-linear measured speedup** up to the physical core count

Find Anything – `github.com/amarz45/find-anything` — September 2026 – Present

- Built a **Java/JavaFX** app and **CLI** that finds local files from **natural-language** queries and answers questions about them, with each cited quote **deterministically verified** against the source text indexed by **Apache Lucene**
- Implemented a **tool-calling agent** with **LangChain4j** that plans searches and reads file excerpts through validated commands, mitigating **prompt injection** by fencing file text in escaped delimiters and keeping all tools read-only
- Isolated the **Apache Tika** parser in a **separate JVM** so malformed files cannot hang or crash the app

ShutterDex – `devpost.com/software/htn-vgd2jp` — **Hack the North 2026 (semifinalist)**

- Led and communicated with a **team of 4** to build a photo-based creature-collecting game in **36 hours**, with a custom ESP32-based camera device and a **FastAPI** server-authoritative battle system over **Wi-Fi**
- Created a **Python** image pipeline: chained Gemini models describe each photo and generate a sprite, and a deterministic processing stage (fixed-palette quantization, outlining) enforces a consistent art style using Pillow

EXPERIENCE

Cybersecurity Analyst (Co-op) — May 2026 – August 2026
Sobeys — Mississauga, Ontario, Canada

- Built a parallel network scanner in PowerShell that classifies devices from parsed Nmap XML output, replacing manual per-host identification and enabling fleet-wide EDR sensor rollout
- Automated CrowdStrike end-of-support reporting in PowerShell using the Falcon **REST API**, correctly classifying ambiguous OS builds (e.g. Windows IoT vs. Enterprise), eliminating a recurring **5+ hour** manual task
- Built an **AI-assisted** workflow analyzing 3 months of API-retrieved email data to identify the top 5 malware families and campaign trends, with findings verified against source data and used in **monthly executive reports**

AI Trainer (Code Reviewer) — June 2025 – September 2025
DataAnnotation — *Remote*

- Authored **unit tests** covering edge cases (off-by-one errors, boundary conditions) and evaluation rubrics defining pass criteria for code-correctness tasks, both used by other reviewers to grade submissions consistently

SKILLS

- **Programming languages:** C++, C, Python, Java, TypeScript/JavaScript, Bash, PowerShell, Rust
- **Technical skills:** Git, GitHub Actions, Linux, Claude Code, GoogleTest, Qt, REST APIs, HTML/CSS, ASan/UBSan
- **Spoken languages:** English (native), French (intermediate)